Multilevel Analysis:

The importance of serving and blocking in a professional male volleyball match

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1 Introduction

In the chapters 42 to 44 the popular high school volleyball manga Haikyuu [2], the team DateTech, nicknamed The Iron Wall, explains how its strategy revolve around disrupting the opponents defence with a good first serve, and then shutting any incoming attack with an efficient block. This focus on serving and blocking is also very present in many training method books [3], since those are actions that all players (except for the libero) will have to perform during a match of volleyball, regardless of their position. However, an interesting question would be to delve into how much of those actions matter at a professional level.

In this report, we will explore this topic using data collected from the Volleyball Nations League (VNL) male group stage during the 2021 season. This data concern individual observations of players performances during matches. This data can be found and collected on their official website. After selecting our variables and treating the data, the dataset contains 1701 observations, with 233 different players belonging to 16 different teams and across 77 matches.

To investigate this subject, we will observe how the serve and block statistics affect the result of a game both with failure and success outcomes. We will also control from "block rebounds", which happens when a blocking player touch the balls, but it doesn't result in a point for either team. This action often reset the game state, so it is expected that this variable will not have an effect on the result of the game. Since we observe the same players for different games, and since those players belong to teams, the use of multilevel modelling is warranted. Finally, we decided to exclude the libero players from this study, since they don't block nor serve.

2 Descriptive Analysis

2.1 Data structure

The three levels at which our data is measured are the teams, the players and the match (or time). Since we measure the individual performances at different points in time, and a greater number of players than matches. On second hand, one important notion to take into account is that players belong to teams in a strict hierarchical structure. We will therefore assume a strict hierarchical structure of matches nested in players, themself nested in teams.

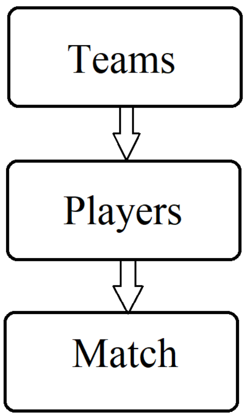


Figure 1: Chosen data structure

This report is mostly interested in the influence of certain variables, and not so much in the specificity of the different countries, we will then chose to first consider our data as panel data.

2.2 Variables

- **id**: (int) Player ID
- **name**: (chr) Player's name **t**: (Date) The date at which this match was played
- **pos**: (chr) the position of the player country: the team to which this player belongs

All the metrics bellow are measured during each game:

- **serve_p**: (num) the number of aces the player scored
- **serve_e**: (num) the number of faults the player made when serving
- **block_s**: (num) the number of points this player scored by blocking
- **block_e**: (num) the number of faults this player made when blocking
- **block_r**: (num) the number of rebound blocks made by this player
- **result**: (binary) 1 if the team of the player won, 0 otherwise

2.3 Distributions

For the summary statistics, we observe:

Variable	Mean	Q1	Median	Q3	[min, max]
serve_p	0.3739	0	0	0	[0,13]
serve_e	1.297	0	1	2	[0,7]
block_s	0.6314	0	0	1	[0,7]
block_e	1.417	0	1	2	[0,12]
block_r	1.038	0	0	2	[0,13]

Table 1: Summary statistics

The distributions are heavily skewed to the left. One explanation may be that those kind of points rarely happen for a single player. Another factor to take into account is that if a played stays on bench during the whole game, his performance will still be registered with a 0, and we have no easy way to tell when it is the case.

If it seems important to model the individual effect of different players, it might not be as obvious that two teams with the same structure would create heterogeneity. We can plot histograms for our different variables in regard to countries, as we did in Annex 1. The overall shapes of distributions for a given variable seem similar for all countries, hinting that the family of distribution is consistent across teams. However, we still observe some important differences, which suggest that the parameters for those distributions might differ, and that the team effect should be part of our models, as those differences could come from very specific over-performing players, but also different strategies from different coaches.

3 Modeling

In all models, the players will be indexed by i, the matches by t and the teams by j. Moreover, the different variables aforementioned except results will be denoted X_k, taking values $X_{k,(it)}$ when observed for an player during a match.

3.1 Fixed Effect Model

The first model to consider is the Fixed Effect (FE) model. Let's suppose that we want to estimate the coefficients β_k in:

$$result_{j,i,t} = \beta_0 + \sum_k \beta_k \cdot X_{k,(it)} + u_{ji} + v_j + e_{jit}$$

After the FE transformation, both unobserved heterogeneity between players u_{ji} and between teams v_j are eliminated, as they are constant regarding time. If we fit a FE model with serve_s, serve_e, block_s, block_e and block_r as covariate, we obtain:

Variable	Estimate	Std. Error	t-value	Pr(>|t|)
serve_p	0.0658003	0.0150874	4.3613	1.384e-05***
serve_e	-0.0074812	0.0097136	-0.7702	0.4413161
block_s	0.0256523	0.0128591	1.9949	0.0462417*
block_e	-0.0270888	0.0081913	-3.3070	0.0009658***
block_r	0.0121421	0.0086274	1.4074	0.1595223

Table 2: FE model

With $R^2 = 0.020312$ and the adjusted $R^2 = -0.13839$

Only 3 of those variable end up being significant at usual levels, being serve_p, block_s and block_e. The most important one is the coefficient for serve_p, with a value of 0.066, showing how much a successful serve can influence the game compared to other ways of scoring. We can also observe that the coefficient for block_s and block_e have a sort of "mirror effect", where they have very close absolute value but opposite sign, respectively 0.026 and 0.027. This suggests that a block leading to a point would grant the same positive effect for the team that scored, no matter which one did.

To see if this model is necessary compared to a simple linear regression, we can use the Breusch-Pagan test. The results are:

$$\text{chisq} = 293.15,\ \text{df} = 1,\ \text{p-value} < 2.2e - 16$$

We can reject the null hypothesis that the variance $\sigma_u^2 = 0$, the use of panel model is then necessary. However, we can still look into Random Effect (RE) models.

3.2 Random effects linear model

Considering at start unobserved heterogeneity between players only:

$$result_{i,t} = \beta_0 + \sum_k \beta_k \cdot X_{k,(it)} + u_i + e_{it}$$

Fitting this model gives us those estimate:

Variable	Estimate	Std. Error	z-value	Pr(>|z|)
(Intercept)	0.5033235	0.0189517	26.5583	< 2.2e-16 ***
serve_p	0.0681594	0.0144823	4.7064	2.521e-06 ***
serve_e	-0.0099570	0.0091436	-1.0890	0.2761702
block_s	0.0297475	0.0124475	2.3898	0.0168561 *
block_e	-0.0290228	0.0079238	-3.6627	0.0002495 ***
block_r	0.0158992	0.0083338	1.9078	0.0564170 .

Table 3: RE model with only unobserved variance for players

The random effect variances are $\sigma^2_{idio} = 0.18812$ and $\sigma^2_u = 0.05523$. For the model in itself, we have $R^2 = 0.025481$ and the adjusted $R^2 = 0.022606$.

Once again, the significant coefficient are same 3, with the same signs. The value for serve_p (0.068) is still over twice as big as the block ones, while the coefficients for block_s (0.030) and block_e (0.029) still have "mirror effects". Those observations help us confirm what we had observed first in the FE model. However, we can go further, and note that the coefficient for the intercept is also significant and much larger than all others (0.5), showing that a difference of one point in serving or blocking does not change drastically the issue of a game.

We can test if this model is preferred to the FE model by performing the Hausman test.

$$\text{chisq} = 5.5095,\ \text{df} = 5,\ \text{p-value} = 0.3569$$

We cannot reject the null hypothesis that the RE model's assumption is to be preferred.

However, we didn't take into consideration the fact that the players belong to teams. According to Baltagi & al. [1], that can be done using a modified Wallace and Hussain (WH) estimator. We can now estimate the model:

$$result_{j,i,t} = \beta_0 + \sum_k \beta_k \cdot X_{k,(it)} + u_{ji} + v_j + e_{jit}$$

The results after fitting this model are:

	Estimate	Std. Error	z-value	Pr(>|z|)
(Intercept)	0.4946117	0.0686285	7.2071	5.716e-13 ***
serve_p	0.0371293	0.0100535	3.6932	0.0002215 ***
serve_e	-0.0076861	0.0053881	-1.4265	0.1537302
block_s	0.0149506	0.0088459	1.6901	0.0910067 .
block_e	-0.0118813	0.0055915	-2.1249	0.0335971 *
block_r	0.0062044	0.0059968	1.0346	0.3008497

Table 4: RE model with nested effect

With $R^2 = 0.010527$ and the adjusted $R^2 = 0.0076081$. However, the assumptions behind this model are very likely not respected, because there is a negative value computed in the variances of random effects:

$\sigma^2_{idio} = 0.18825$, $\sigma^2_u = -0.01857$ and $\sigma^2_v = 0.07930$ We also established that changing the estimation method to swar or amemiya did not solve this problem, nor adding time independant variables.

Nevertheless, it might still be interesting to note that the same variables have significant coefficients will similar signs as in previous model, thought we will not risk any further analysis there.

3.3 Logistic model

Another way we can look at this problem is to try to modelise the probability of winning with a logistic mixed model. Let $result_{jit}$ follow a Bernouilli distribution of parameter π_{jit}. With the chosen data structure, we can estimate the model:

$$log(\frac{\pi_{ij}}{1-\pi_{ij}}) = \tilde{\beta_0} + \sum_k \tilde{\beta_k} \cdot X_{k,(it)} + u_{ji} + v_j + e_{jit}$$

After the fit we observe the estimated values:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-0.03004	0.35661	-0.084	0.9329
serve_p	0.37962	0.08709	4.359	1.31e-05 ***
serve_e	-0.05162	0.04583	-1.126	0.2600
block_s	0.14245	0.06702	2.126	0.0335 *
block_e	-0.13722	0.04237	-3.239	0.0012 **
block_r	0.05227	0.04457	1.173	0.2409

Table 5: Logistic mixed model

With $AIC = 1823.9658$, $BIC = 1867.4775$ and a log-likelihood of -903.9829.

The estimated variances for the random effects are $\sigma^2_e = 0.18825$, $\sigma^2_u = 1.322e-37$ and $\sigma^2_v = 1.924$ (as we are working with generalised mixed models, the residual variance is fixed to $\frac{\pi^2}{3}$). We observe a huge variance in the country level random effect, and a variance almost null for the individual level random effect. This result is surprising considering that the profile of players can vary a lot within a specific team, especially since they have different positions.
We observe 3 significative estimates. The coefficient for serve_p is once again the most important one with a positive value of 0.38, confirming its relevance. We also observe the same "mirror effect" as previously, with both block_s and block_p having absolute values of 0.14.

Now that we fitted a logistic model, we can get a better idea on how the different variables affect the probability of winning, we can compute the Average Marginal Effects (AME):

$$AME_k = \frac{1}{N} \sum_{j,i,t} \frac{\partial \pi_{jit}}{\partial X_{k,jit}}(X_{jit})$$

We can compute the partial derivative:

$$\frac{\partial \pi_{jit}}{\partial X_{k,jit}} = \tilde{\beta_k} \frac{exp(\tilde{\beta_0} + \sum_k \tilde{\beta_k} \cdot X_{k,(it)} + u_{ji} + v_j + e_{jit})}{(1 + exp(\tilde{\beta_0} + \sum_k \tilde{\beta_k} \cdot X_{k,(it)} + u_{ji} + v_j + e_{jit}))^2}$$

We observe for the significative variables the following AMEs:

serve_p	block_s	block_e
0.06507	0.02442	-0.02352

Those value represent the average change in the probability of winning corresponding when a point in made this way. On average, scoring on the serve increase the odds of winning by 6.5%, while a point scored (or conceited) changes the odds by around 2.4%.

4 Conclusion

4.1 Discussion

The results found above and summarised bellow should be taken with many precautions. One flaw previously mentioned is the presence of players sitting on the bench in the dataset, and the impossibility of distinguishing them from a player that would take non of those actions on the field. An important factor that was also not considered in our study is that the effect of a point scored during a match will greatly change depending on when it happens. Series of consecutive successful serves are known to hurt the moral of defending players, and often lead to time-outs by coaches. On the other hand, a successful block can help kill the mental momentum gained by the attacking team. Since our data is aggregated to match level, such dimensions could not be considered.

4.2 Findings

The most important variable in our models was consistently the number of points scored on serves per match, and it's coefficient was always at least twice as big as the others. We can also note that the number of serves failed was not significant in any model. One interpretation may be that failing on the serve happens when someone tries something risky, that could also lead to a point as well. Those players also possess a better ability to stay focused, even after making such faults, which could negate it's impact.

The two coefficients for the variables regarding points scored after a block also had almost equal absolute values and opposite in all our models. This suggest that blocking might be a double-edge sword in front of a professional attacker that can push the blocker to the fault.

5 Literature

References

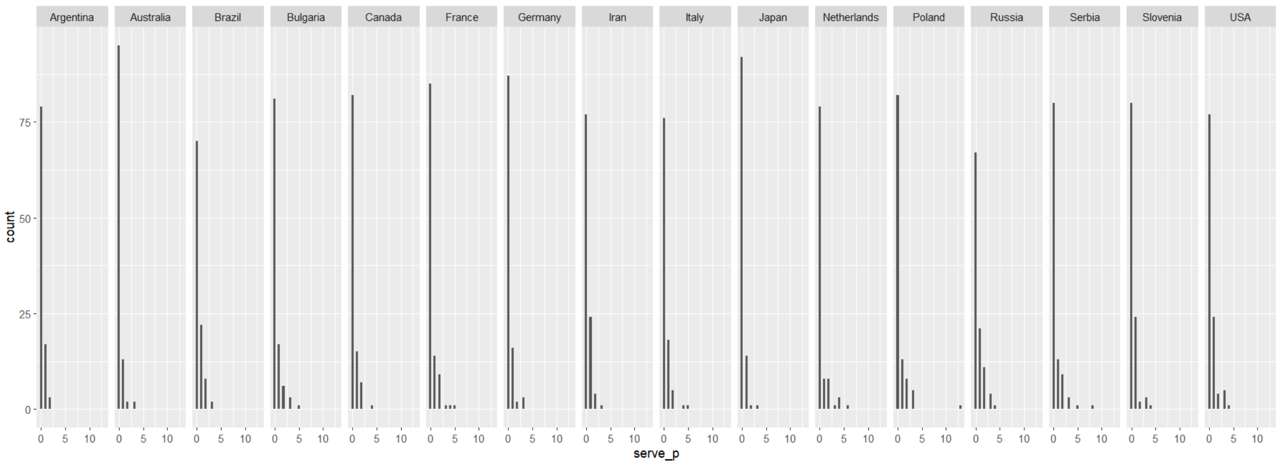
[1] Seuck Jung Byoung Baltagi, Badi Song. The unbalanced nested error component regression model. *Journal of Econometrics*, 2001.

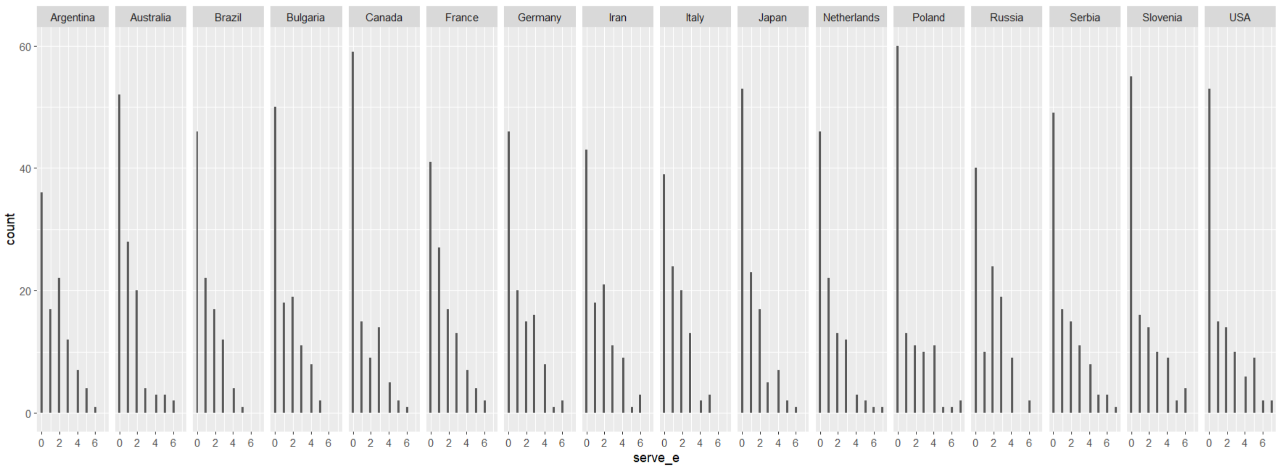
[2] Haruichi Furudate. *Haikyuu*. 2013.

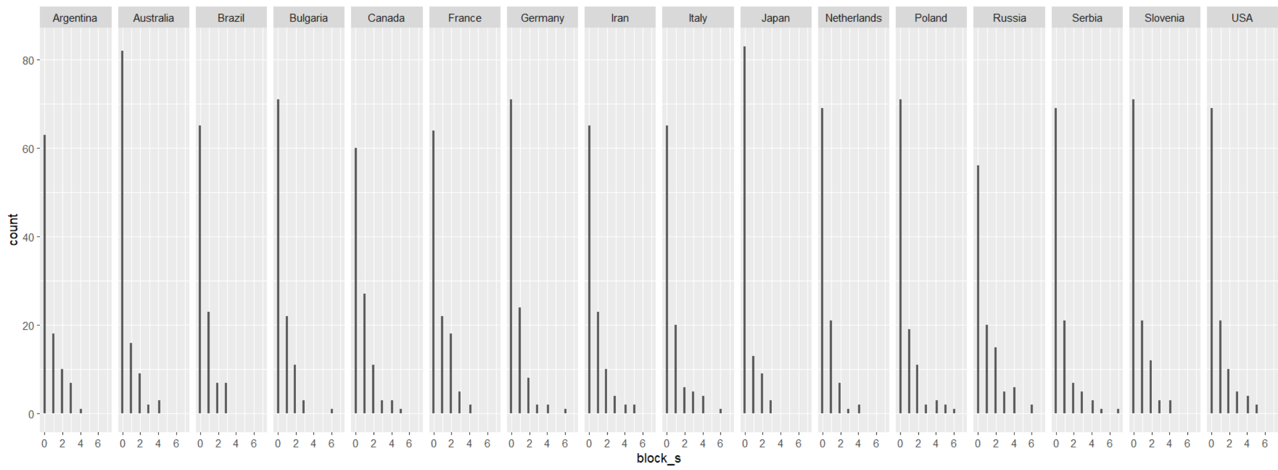
[3] Stéphane Lamache. *Volleyball. Méthode d'entraînement*. 2003.

6 Annex

6.1 Plots for the distributions per country



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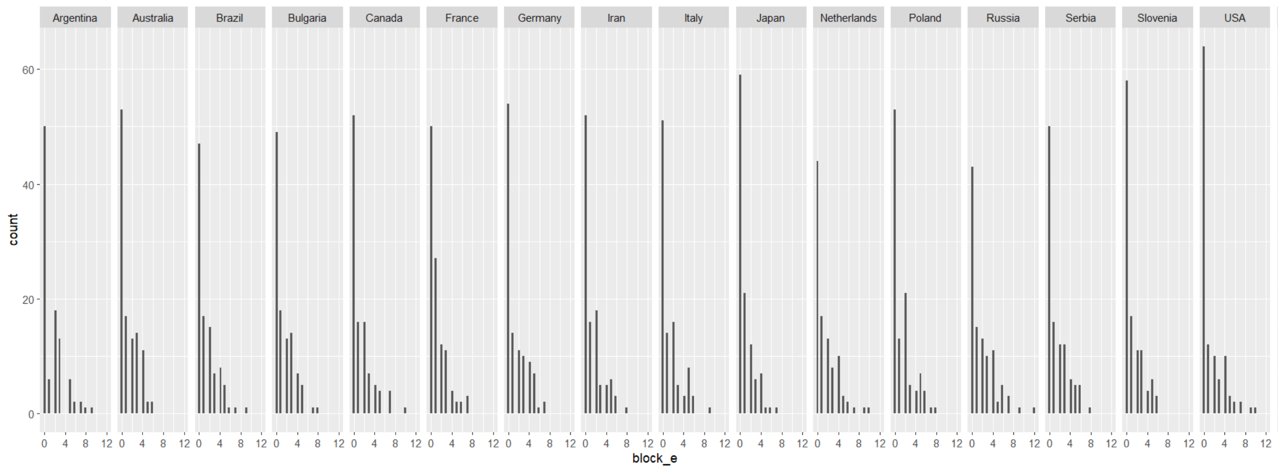
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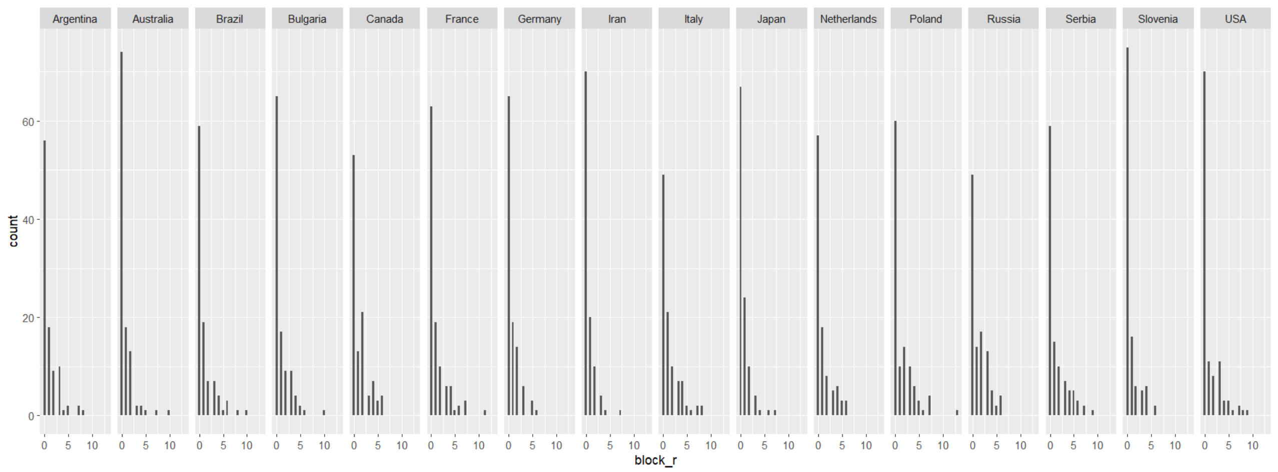
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